



Stocks Opening Price Gaps and Adjustments to New Information

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Accepted: 29 January 2023

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Abstract

This research studies different gap opening price strategies using artificial intelligence and big data analysis to learn how fast new information is absorbed into the stock's price. Our system is designed to optimize trading results of different gap opening investment strategies. Our data consist of ten years of daily trading prices of all the stocks comprising the three major U.S. stocks indices: S&P 500, Nasdaq100, and Russell 2000. The scope of this research, to the best of our knowledge, has never been attempted before, covering most of the U.S.A. economy across various economic conditions and market trends. We found that negative gap openings are much greater than positive gaps opening. This result is stronger for Russell2000 stocks and Nasdaq100 stocks than for S&P500 stocks. Moreover, consistent with the theoretical framework, the price adjustment for bad news was found to be quicker than for good news. We also found that after positive gaps opening price drifts occur, the stock's price rises even stronger, providing profitable trading opportunities.

Keywords Price Gaps · Stocks Information · Long Short Strategies · Algorithmic Trading · Bad and Good News

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1 Introduction

A gap is a change in price levels of a financial asset between the close and open of two consecutive days. Most gaps are formed as a direct result of an external shock that occurred during the time the stock market was closed. An external shock can be positive or negative and can influence a single stock, sector, or the entire economy. Gaps are created because of an imbalance between demand and supply at the trade opening bell. A gap up is driven by aggressive buyers relative to a lean supply at the prior day's closing price. A gap down is created because of aggressive sellers relative to buyers at the prior day's closing price. There are two kinds of gaps, full and partial. A full gap up occurs when the opening price is greater than yesterday's high price and a full gap down occurs when the opening price is less than yesterday's low. A partial gap up occurs when today's opening price is higher than yesterday's close, but not higher than yesterday's high. A partial gap down occurs when the opening price is below yesterday's close, but not below yesterday's low. The size of the gap and the price drift during the daily trading period is driven by investor's psychological state of mind that drives the price level to overshoot up or down more than it should according to economic equilibrium valuation. In a perfect market, as described by Fama (1970), stock prices reflect all new information about firms immediately and completely. In such a market, the overnight new information should be absorbed into the stock price immediately with the opening bell, hence, price gaps up or down should reflect the new information, with no substantial price drifts during the trading day, provided that no additional information was advertised. Recent studies have found that price drifts do exist, and that it takes time to reach price equilibrium that objectively reflects the economic news. For example, Savor (2012) showed that events of extreme price shocks, accompanied by information, are followed by drift within a three-month period. Cheng et al. (2019) suggested that stock price continuation is stronger when investors are more optimistic about the firm's prospects. They also found that following extreme price events, the stock price continues to drift when the event is associated with new public information, and that such continuation is stronger for firms with lower past information surprises.

Negativity bias has been the focus of attention in psychology since Kahneman and Tversky published their prospect theory (Kahneman & Tversky, 1979; Tversky & Kahneman, 1991). The prospect theory suggests that people prefer to avoid losses over acquiring gains, a phenomenon known as loss aversion. A large body of research indicates that negative information has a processing advantage over positive information because processing negative information enables avoiding potential dangers (for example Baumeister et al., 2001). Moreover, negative information is detected faster than positive information (Dijksterhuis & Aarts, 2003), and exerts a greater influence on impression formation (Peeters & Czapinski, 1990). Fessler et al. (2014) found that urban legends typically contain three times as much information about hazards as they do about benefits. Such findings are consistent with the idea that the social transmission of information is distorted by a bias that favors negative informational elements. Rozin and Royzman (2001) described those negative entities are stronger than the equivalent positive entities and that the negativity of negative events grows more rapidly as they are approached in space or time than does the

positivity of positive events. Like Rozin and Royzman, we also find that negative information that was published when the market is closed is captured almost fully into the stock price when the market opens with a negative price gap down. On the other hand, positive information is not absorbed instantly into the stock price when a positive gap up appears. In such a case, a positive drift occurs during the trading day. Baumeister et al. (2001) argued that in most situations, negative events will produce larger, more intense consequences than positive events of comparable magnitude. We also documented inequality in reaction to bad and good news. The bad news results in a substantial immediate dip down of the stock price, while good news results in a more modest up drive of the stock price that may be concluded eventually in one or more trading days. Our data consisted of ten years (from the beginning of January 2010 till the end of December 2019) daily open-close, high-low prices of S&P500, Nasdaq100, and Russell 2000 stocks. The observed total number is 2,600 different-sized stocks covering all sectors of the U.S.A. economy. Moreover, the ten-year data contain all stocks market trends including up down and seesaw. Such a large sample has rarely been used in previous financial research. The following results have been obtained: 1. A negative opening gap is much greater in size than a positive opening gap. 2. This result is stronger for Russell2000 stocks and Nasdaq100 stocks than for S&P500 stocks. 3. After a positive opening gap, a price drift occurs, pushing the stock's price up, and providing profitable trading opportunities. 4. A change in the direction of the opening gap from positive to negative, and vice versa, in two consecutive days results in a stronger price drift on the trading day than in two same directions.

2 Literature Review

A growing body of financial literature has dealt with the issue of adjustment rate of stocks to both corporate and non-corporate news. Machmuddah et al. (2020) observed stock prices of customer goods before and after the COVID-19 pandemic and found a significant difference between the daily closing stock price and the volume of stock trade before and after the COVID-19 pandemic. Moreover, they reached the conclusion that the more complete the provided information, the more efficient the market. Zhang et al. (2022) examined the rate that stocks absorbed Covid 19 shocks and found that the absorption rate of the non-cyclical industries such as utilities and consumer staples is high, while the cyclical industries such as banking, real estate, and energy have lower absorptive rate. Tetlock (2014) concluded that shocks are not perfectly absorbed the right way. However, the shock's effect is ambiguous; some studies find return reversal, while others find drifts. Frank and Sanati (2018) found differences in market responses to positive and negative new information. They observed that when there is a news story about a firm, positive price shocks are followed by a reversal, while negative ones result in drift. This is interpreted as the stock market overreaction to good news and underreaction to bad news. Moreover, they found that there is a stark difference in market responses to positive and negative news shocks. On a news day for a particular stock, the price of a stock is typically overreacting to good news and underreacting to bad news. Akhtar

et al. (2011) examined the equity market reaction to the monthly release of Australian consumer sentiment news. Their results indicate that consumer sentiment has valuable information content. Further, they documented a version of the “negativity effect” in which, upon announcement of bad (good) sentiment news, the equity market experiences a significant negative announcement day effect. They also found that the market recovers from the bad news shock relatively quickly post announcement. Cheng et al. (2019) argued that when an information-based price movement occurs; investors combine the new information with their private information to adjust their expectation of the stock return. Because not all investors have precise knowledge of the firm and their levels of attention are different, therefore, the observed magnitude of price continuation should vary with the level of investor attention and surprise. Other research has also documented that investor react to a public release of information with different time lags and magnitudes depending on the level of information asymmetry (Zhang 2006, Huang 2012, Huang & Cheng 2013). Their research has established that investors make informed trading decisions according to the extreme shocks and that stock prices would incorporate the continuous reflection of the news. Savor (2012) argued that information-based price continuation is more significant than price continuation which is unrelated to the recent release of public information. Savor (2012) and Govindaraj et al. (2014) found that underreaction to information shocks led to drifts in returns over the next two to three months. Stock adoption of new information depends heavily on corporate managerial disclosure. New information can become public at once or in a sequence of investor relations events. Chapman et al. (2019) examine whether managers can reduce the detrimental effects of information overload by spreading out disclosures. They found that managers are more likely to engage in disclosure spreading when the new information is complex, uncertainty is high, and disclosure news is more positive. Their interesting conclusion can explain our finding of a substantial immediate price decline in response to bad news, while the good news is followed by a lower upside effect followed by a drift until the end of the trading day.

3 Research Design and Data

In this research, we examine the price behavior of stocks prices after gap opening for both directions up and down. Gaps opening usually follow a piece of new information about the firm, the industry, or the entire economy enabling us to study how fast the new information is absorbed into the stock’s price. We base our study on ten years of daily data of all S&P 500, Nasdaq 100, and Russell 2000 stocks. The scope of our data, both in the time frame and in the variety of companies, is unique in the financial research. Our aim is to learn about opening gaps and the movement of the stock price after a gap opening, in order to understand if positive or negative price opening gaps contain all the information that was published before the opening bell. If the market adjusts the stock’s price immediately once the market opens, the closing price should be close to the opening price if no new information was published during the trading day. However, if the closing price strongly drifts from the opening price, traders can exploit the market inefficiency. We measure and compare the opening gaps sizes and

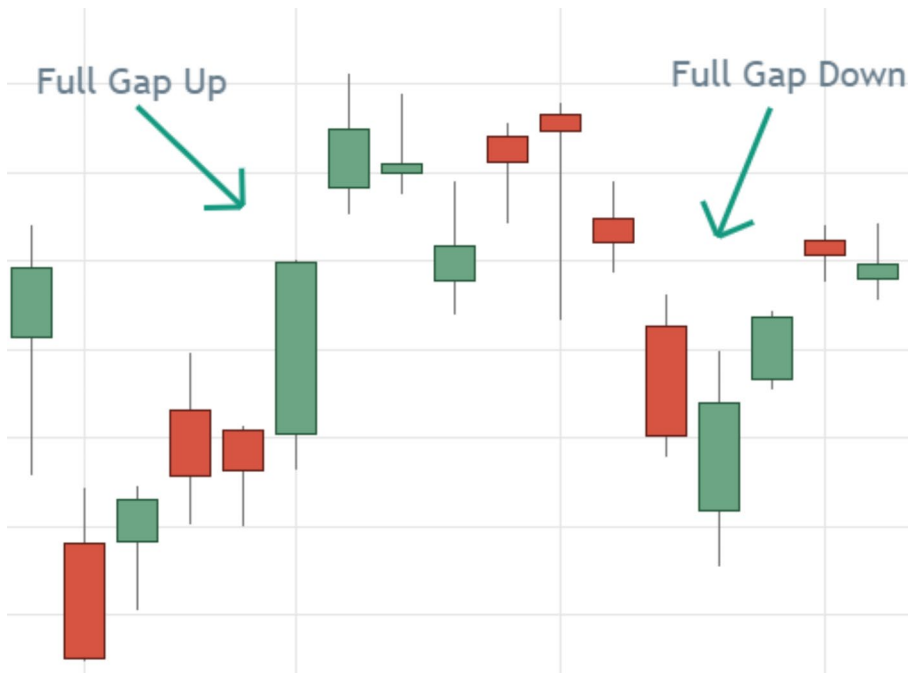


Fig. 1 Full Gaps Up and Down. (Notes: a green bar means that the daily closing price is higher than the opening price. A red bar means that the daily closing price is lower than the opening price. The shadows above and below the bars symbolize the highest and lowest daily price)

the daily drifts for both positive and negative openings to learn about the difference of market efficiencies for both good and bad news, and the possibility to take advantage of the daily drifts by traders. We also constructed trading strategies that are based on two consecutive gaps opening to analyze price adjustment over more than one day.

There are two kinds of opening gaps: A full gap-up (down) means that the opening price of day 1 is higher (lower) than the highest (lowest) price at day 0. Figure 1 shows an example of full gaps.

A partial gap up (down) meaning that the opening price on day 1 is higher (lower) than the closing price of day before but not higher (lower) than the highest (lowest) prices at day 0 as demonstrated in Fig. 2.

The comparison of gaps up with gaps down strategies enables us to assess the economic impact of good and bad news published before the opening bell, on the stock's price. The opening gap size captures the impact of the news published before the opening bell, while the length of the daily candlestick captures the drift of that information during the entire trading day. The length of the candlestick represents the difference between the opening and the closing daily price. The longer the candlestick is, the higher the daily price drift occurs, meaning that the opening gap failed to capture the entire new information published before the opening bell and that a trader can take advantage of the daily price drift. Behavioral finance literature (see for example Rozin & Royzman, 2001) predicts that negative information shocks will be greater than positive shocks. Our unique methodology enables us to measure the

Fig. 2 Partial Gaps Up and Down. (Notes: a green bar means that the daily closing price is higher than the opening price. A red bar means that the daily closing price is lower than the opening price. The shadows above and below the bars symbolize the highest and lowest daily price)



size of the gap opening responsiveness to positive or negative information and to document daily price drifts that are followed gap opening. The analysis of the data was carried out by using R coding. The algorithm buys the stock when it finds a proper gap and sells at the end of the trading day. For each stock, we find the number of trades, the distribution of win (loss) days, the average percentage of a daily win (loss), net daily gain, and maximum daily win (loss) according to the opening gap. The results are presented in Section 4: first long and short single-day gap formation for our three stocks samples (S&P500, Nasdaq100, and Russell 2000). Following, we present the results of gap openings after two sequential gaps openings.

4 Results

We start our analysis by looking at the performance of trading according to the long gap opening of our three samples. Long daily opening results from a piece of good new information that concerns the individual stock, the sector to which the stock belongs, or even the entire economy. The average opening gaps in the case of “efficient market” should contain all known information at the opening time. A drift occurs if the price continues to move forward during the trading session, insinuating that the new information did not absorb the stock price at opening time. A drift can be taken advantage of by traders that buy the stock at the opening bell and sell it at the end of the trading session. Our aim is to investigate if different behavior can be identified between our samples. In addition, to ensure that the obtained results are statistically significant and robust, we performed three statistical tests for Net Daily Gain and the proportion of Win Days within each strategy. Specifically, we run one-sided one-sample t-tests, sign-test, and Wilcoxon signed-rank tests to check whether the Net Daily Gain is greater than zero and whether the proportion of Win days is greater than 50%. The reason we perform three different tests for every net daily gain and the win days’ proportion within each strategy is to ensure that the results are not dependent on specific statistical assumptions on the data (such as normality or symmetry) and, thus robust. The test results are consistent with each other and support

Table 1 Long Trading Results of Partial and Full Positive Gaps for S&P500, Nasdaq100 and Russell 2000 Stocks

Strategy	Statistic	Gap (%)	Trading days	Win days (%)	Loss days (%)	Average daily Win (%)	Average daily Loss (%)	Net daily Gain (%)	Max daily Win (%)	Max daily Loss (%)
S&P 500										
1	Average	0.87	733	55	45	1.26	-1.16	0.16	7.25	-6.64
	Median	0.20	796	53	47	1.04	-0.99	0.09	6.60	-6.14
2	Average	1.10	212	58	42	1.41	-1.23	0.30	5.79	-5.64
	Median	0.10	249	55	45	1.12	-1.04	0.15	5.06	-5.22
Nasdaq 100										
1	Average	1.92	455	55	45	1.79	-1.51	0.31	7.63	-6.45
	Median	0.70	272	54	46	1.34	-1.18	0.18	6.66	-5.79
2	Average	1.51	172	60	40	1.91	-1.62	0.50	6.65	-5.91
	Median	0.40	132	56	44	1.35	-1.26	0.20	5.66	-5.50
Russell 2000										
1	Average	4.0	257	50	50	4.32	-4.16	0.16	15.48	-11.41
	Median	2.6	30	52	48	2.38	-2.39	0.09	11.27	-9.65
2	Average	3.5	73	54	46	4.80	-4.36	0.58	14.03	-10.33
	Median	2.2	13	54	46	2.52	-2.26	0.32	10.12	-8.41

Notes=Strategy 1 positive partial gap, Strategy 2 positive full gap, Net daily Gain=Average daily Win*Win days (%) + Average daily Loss*Loss days (%).

our main findings (see Appendix A). Table 1 shows long trading results based on a positive partial gap (strategy 1) and a full positive gap (strategy 2).

Table 1 shows that gaps opening are much larger for small stocks (Russell 2000) than for larger stocks. Moreover, the gaps are higher for the Nasdaq100 stocks, which contain technological stocks, than for the S&P500 industrial stocks. Those results are consistent with the known fact that small stocks tend to be more volatile than big stocks, and technology stocks are more volatile than industrial-utilities stocks. These results are consistent for the partial and full gaps strategies. The difference in the stock's volatility is also represented by the maximum daily win and loss of the different samples. Table 1 also reveals that the net daily gain for all three samples is larger for the full gap strategy than for the partial gap. Full positive gaps occur when the opening price is detached entirely from the last trading day, symbolizing that major information has been added to the market that strongly pushes the stock price forward. The highest net daily gain occurred to the Russell 2000 stocks followed by the Nasdaq100 stocks and last for the S&P500 stocks. These results are significant since these strong drifts mean that it takes the entire trading session for the stock's price to adjust to the new positive information. Moreover, a trader can take advantage of this price drift by identifying positive full gaps opening, buying the stock, and selling it at the trading session end. By doing so, the trader will, on average, get a positive gain of 0.58, 0.50, and 0.30 percent for the Russell2000, Nasdaq100, and S&P500 stocks, respectively. The highest winning trade for the full gap strategy was 60 percent for the Nasdaq100 stocks, followed by 58 percent for the S&P500 and 55 percent for the Russell 2000 stocks. Table 1 also shows that, for the partial gaps trading strategy (strategy 1), the price drift during the daily trading session is much narrower than

Table 2 Short Trading Results of Partial and Full Negative Gaps for S&P500, Nasdaq100 and Russell 2000 Stocks

Strategy	Statistic	Gap (%)	Trading days	Win days (%)	Loss days (%)	Average daily Win (%)	Average daily Loss (%)	Net daily Gain (%)	Max daily Win (%)	Max daily Loss (%)
S&P 500										
3	Average	2.58	348	52	48	1.92	-1.73	0.18	7.75	-5.95
	Median	1.00	120	51	49	1.46	-1.22	0.15	7.13	-5.76
4	Average	2.78	106	55	45	2.14	-1.80	0.38	6.78	-5.56
	Median	1.10	40	53	47	1.54	-1.27	0.20	6.11	-5.38
Nasdaq 100										
3	Average	4.13	187	49	51	2.59	-2.53	0.02-	7.61	-6.01
	Median	3.90	12	51	49	2.09	-1.66	0.25	6.87	-5.74
4	Average	4.35	39	54	46	2.82	-2.70	0.28	6.61	-5.71
	Median	4.40	6	52	48	2.39	-2.06	0.25	5.28	-5.25
Russell 2000										
3	Average	4.1	256	49	51	4.12	-4.28	0.16-	17.18	-11.83
	Median	3.2	18	51	49	2.58	-2.54	0.07	13.29	-10.36
4	Average	4.0	51	53	47	4.93	-4.80	0.36	15.19	-10.48
	Median	3.7	7	53	47	3.09	-2.95	0.25	11.43	-9.04

Notes=Strategy 3 negative partial gap, Strategy 4 negative full gap, Net daily Gain=Average daily Win*Win days (%) + Average daily Loss*Loss days (%).

the drift for the full gap strategy (strategy 2). This phenomenon is consistent for all three sample stocks when the highest daily gain documented was 0.3 percent for the Nasdaq100 stocks. Table 2 summarizes the results for short trading results for partial and full negative price gaps.

Table 2 shows that the negative gaps are stronger than positive gaps for both partial and full gaps and for all three samples. The Nasdaq100 stocks average full gap down is 4.35 percent, compared to the full gap rise of only 1.51 percent presented in Table 1. This is also true for the S&P500 stocks, for which the average full gap down is 2.78 percent, compared to the average rise of only 1.1 percent on an uptrend opening. This is also consistent for the Russell200 stocks; however, the difference between up and down gaps is much smaller. This observed phenomenon is consistent with the theoretical framework in the literature, advocating for a stronger investor's reaction to bad news than for good news (Rozin and Royzman 2001; Fessler et al. 2014). Another important result is the drop in the net gains (full and partial gaps) for the Nasdaq100 and Russell200 stocks for positive and negative gaps. The only stocks that experienced a slight rise in net gain were the S&P500 stocks. These results indicate that the market is more efficient for bad news than for good news, absorbing bad news more quickly into the stock's price than good news. Moreover, small, or non-price drift was detected for the Nasdaq100 and Russell2000 stocks. These results are very important since they indicate that the market better evaluates companies' economic influence of bad news on the stock price, rather than good news before the opening bell. These results also indicate that long buying after a positive opening gap can be a profitable strategy for traders, much more than short selling a stock after a negative gap opening of Nasdaq100 and Russell2000 stocks. For the S&P500, simi-

lar drifts were detected on positive and negative gap opening. The Nasdaq100 and Russell2000 stocks movement after bad news demonstrates high market efficiency for those stocks, and leaves no room for traders to take advantage of the daily drift.

We now report in Table 3 the trading results after two sequential days gaps opening signaling before the actual trading day. Strategy 5 consists of two negative partial gaps (t-2 and t-1) followed by a positive gap on the buying day t. Strategy 6 involves a negative partial gap at t-1 and a positive gap on the buying day t. Strategy 7 posits a positive partial gap at t-1 and a positive gap on the buying day t. The purpose of the two days gaps trading analysis is to identify longer than one-day price drifts and information stock absorption. More than one gap opening identifies strong investors' emotional reactions to new information. The information is usually introduced to the stock market at once, according to the Securities and Exchange Commission (SEC) regulations. However, the interpretation of the news or additional news might take longer to reach the financial markets.

Strategy number 5 identify two days of sequential negative gaps opening followed by a positive gap opening. This situation appears when investors estimate that the two previous days' downtrends have pushed prices down too much and must be aggressively corrected. Table 3 demonstrates that the drift on the third day up gap opening, followed by two previous days negative gap opening, is very large for the Russell2000 stocks, resulting in 1.16 percent net gain and 67 percent winning days. This phenomenon is also strong for the Nasdaq100 and the S&P500 stocks, 0.71 and 0.59 percent, respectively. The change in the trend of Russell2000 and Nasdaq100 stocks appears after the large gap dive documented in Table 2. Such substantial gains provide a fertile ground for algo traders, who can set their algorithm to identify two sequential negative gaps opening, followed by a positive gap opening, and buy the stock on that positive opening day.

Strategy number 6 comprised of one negative gap opening followed by a positive gap opening on the following day. Again, the Russell2000 stocks exhibit a high net gain of 0.87 percent followed by 0.75 percent for the Nasdaq100 stocks and 0.37 percent for the S&P500 stocks. Such large positive drifts posit that opening positive gaps fail again to capture the entire known information that can be taken advantage of by traders. Strategy number 7 is based on two consecutive days with positive price gaps. This strategy, as do the other two strategies, facilitates positive net gains of 0.58, 0.33, and 0.21 percent for the Russell2000, Nasdaq100, and S&P500 stocks, respectively. These documentations are proving that a positive gap price drift is stronger after more than one sequential gap opening, regardless of their direction, than a single day gap price drift, enabling traders to achieve high gains that are due to market inefficiency to quickly absorb good news. Moreover, a change in the direction of the price gap opening from negative to positive has resulted in a stronger price drift on the trading day than after two consecutive days of a positive gap opening. Table 4 shows short trading results, which are based on more than one-day sequential gap opening.

Strategy number 8 identifies two days of sequential positive gaps opening followed by a negative gap opening. As for the long positions summarized in Table 3, the short trade strategy that follows two sequential positive gaps opening has obtained substantial gains for a single day of trade. The Russell2000 again leads the gains with

Table 3 Long Trading Strategies Based on Previous Days Signaling for S&P500, Nasdaq100 and Russell 2000 Stocks

Strategy	Statistic	Gap t-1 (%)	Gap t (%)	Trading days	Win days (%)	Loss days (%)	Average daily Win (%)	Average daily Loss (%)	Net Daily Gain (%)	Max daily win (%)	Max daily loss (%)
S&P 500											
5	Average	0.89	0.14	61	64	36	1.43	-0.91	0.59	4.81	-2.87
	Median	0.40	0.00	55	60	40	1.17	-0.84	0.37	4.32	-2.70
6	Average	0.92	0.25	207	59	41	1.39	-1.10	0.37	6.04	-4.61
	Median	0.30	0.10	186	56	44	1.12	-0.95	0.21	5.53	-4.39
7	Average	0.65	0.49	428	57	43	1.23	-1.13	0.21	6.39	-6.00
	Median	0.20	0.10	426	55	45	1.06	-0.97	0.14	5.56	-5.43
Nasdaq 100											
5	Average	0.98	0.29	53	66	34	1.60	-1.02	0.71	4.72	-3.07
	Median	0.40	0.10	40	60	40	1.38	-0.94	0.21	4.21	-2.82
6	Average	1.45	0.52	126	64	36	1.92	-1.34	0.75	6.13	-4.59
	Median	0.70	0.30	49	58	42	1.54	-1.18	0.40	5.17	-4.21
7	Average	1.28	0.76	349	57	43	1.65	-1.41	0.33	7.31	-6.42
	Median	0.40	0.20	319	54	46	1.34	-1.17	0.22	6.36	-5.76
Russell 2000											
5	Average	1.8	0.2	47	67	33	2.51	-1.60	1.16	8.52	-4.42
	Median	1.1	0.0	31	63	37	1.96	-1.32	0.75	6.47	-3.78
6	Average	1.9	0.5	136	62	38	2.58	-1.95	0.87	10.81	-6.84
	Median	1.1	0.1	77	58	42	1.95	-1.55	0.50	8.43	-5.90
7	Average	3.0	2.0	157	56	44	3.86	-3.54	0.58	14.96	-10.58
	Median	1.6	0.5	54	54	46	2.37	-2.08	0.34	10.61	-8.81

Notes: Strategy 5- two consecutive negative partial gaps (t-2 and t-1) followed by a positive gap on the buying day t. Strategy 6-negative partial gap at t-1 and positive gap on the buying day t. Strategy 7- positive partial gap at t-1 and a positive gap on the buying day t, Net daily Gain=Average daily Win*Win days (%) + Average daily Loss*Loss days (%).

Table 4 Short Trading Strategies Based on Previous Days Signaling for S&P500, Nasdaq100 and Russell 2000 Stocks

Strategy	Statistic	Gap t-1 (%)	Gap t (%)	Trading days	Win days (%)	Loss days (%)	Average daily Win (%)	Average daily Loss (%)	Net Daily Gain (%)	Max daily win (%)	Max daily loss (%)
S&P 500											
8	Average	0.78	0.19	47	63	37	1.42	-0.96	0.54	4.75	-2.88
	Median	0.50	0.10	43	60	40	1.26	-0.88	0.40	4.23	-2.68
9	Average	0.82	0.33	151	59	41	1.34	-1.02	0.37	5.50	-3.94
	Median	0.50	0.10	110	55	45	1.15	-0.93	0.21	4.82	-3.60
10	Average	1.72	1.26	208	56	44	1.84	-1.55	0.35	7.50	-5.93
	Median	0.80	0.20	144	53	47	1.52	-1.23	0.22	6.69	-5.71
Nasdaq 100											
8	Average	1.02	0.19	40	64	36	1.59	-1.06	0.63	4.89	-2.96
	Median	0.50	0.00	32	60	40	1.45	-0.97	0.48	4.36	-2.68
9	Average	1.14	0.33	115	58	42	1.52	-1.20	0.38	5.58	-4.27
	Median	0.80	0.10	86	55	45	1.32	-1.07	0.24	5.03	-3.73
10	Average	2.38	1.64	132	55	45	2.26	-1.96	0.36	7.76	-6.24
	Median	1.60	0.40	49	52	48	1.95	-1.63	0.23	6.95	-5.74
Russell 2000											
8	Average	1.6	0.2	46	67	33	2.30	-1.58	1.04	8.00	-4.59
	Median	1.1	0.0	34	64	36	1.98	-1.34	0.79	6.44	-3.71
9	Average	1.6	0.3	151	62	38	2.30	-1.86	0.70	10.87	-7.01
	Median	1.0	0.0	104	58	42	1.91	-1.53	0.46	8.37	-5.81
10	Average	3.1	2.3	130	55	45	3.78	-3.85	0.33	15.49	-10.92
	Median	1.7	0.5	32	54	46	2.71	-2.43	0.35	11.97	-9.38

Notes: Strategy 8- two consecutive positive partial gaps (t-2 and t-1) followed by a negative gap on the shorting day t. Strategy 9- positive partial gap at t-1 and a negative gap on the shorting day t. Strategy 10- negative partial gap at t-1 and a negative gap on the shorting day t. Net daily Gain=Average daily Win*Win days (%) + Average daily Loss*Loss days (%).

1.4 percent, followed by 0.63 percent for nasdaq100 stocks and 0.54 percent for the S&P500 stocks with an impressive percentage of winning trades (67,64 and 63 percent). These results are lower than those of strategy number 5, proving that price drifts followed by negative opening gaps are lower than price drifts after a positive opening gap. However, the daily gains are still substantially high and can be taken advantage of by traders.

Strategy number 9 captures a one-day positive gap that follows a negative opening gap the next day. Shorting the stock according to this strategy would have gained 0.7, 0.38, and 0.37 percent for the Russell2000, Nasdaq100, and S&P500 stocks, respectively. Again strategy 9 is less effective than strategy 6 that documented stronger price drift on a positive gap opening than on a negative price gap opening for the Russell2000 and Nasdaq100 stocks. Strategy number 10 is comprised of two negative consecutive days of price opening. This strategy has produced the lowest net gains for investors. 0.33, 0.36, and 0.35 percent for the Russell2000, Nasdaq100, and the S&P500 stocks. While this short strategy is less effective than the similar long strategy (strategy 7) for the Russell2000 stocks, it was found to be more effective for the larger stocks (Nasdaq100 and S&P500). The comparison of the results summarized in Table 3 to those summarized in Table 4 reveals that two days of signaling before the trading day provide a substantial gain for long and short traders. This finding shows that turbulence in stock prices enhances daily drift that can be taken advantage of by algo traders. Moreover, for most examined strategies, the price drifts after a positive opening gap was higher than after a negative opening gap, proving again that bad news is absorbed faster into the stock price than good news. Again, a change in the direction of the price gap opening from positive to negative has resulted in a stronger price drift on the trading day than after two days in a row of a negative gap opening.

5 Summary and Conclusions

This research studies different gap opening price strategies, to learn how quickly new information is absorbed into the stock's price. Our data consists of ten years of daily trading of all the stocks that comprised the three major U.S. stocks indices: S&P500, Nasdaq100, and Russell2000. The scope of this research, to the best of our knowledge, has never been attempted before, covering most of the U.S. economy across various economic conditions and market trends. We found that negative openings gaps are much greater than a positive opening gaps. This result is stronger for Russell2000 stocks and Nasdaq100 stocks than for S&P 500 stocks. Moreover, consistently with the theoretical framework, the price adjustment for bad news was found to be quicker than for good news. We also found that after a positive gap, price drifts occur that make the stock's price rise even higher, providing profitable trading opportunities. Furthermore, gap signaling is stronger if it occurs on two consecutive days, and it facilitates a stronger price drift on the trading day. Finally, a change in the direction of the opening gap from positive to negative, and vice versa, in two consecutive trading days results in a stronger price drift than two days with the same directions of an opening gaps.

Our results are essential for understanding the effect of human behavior on stock prices. As for other behavioral aspects, we also find that the reaction to negative information is observed quicker in the stock's price than to positive information. This phenomenon can be taken advantage of by robotic or human traders that can exploit these market inefficiencies. Furthermore, the realization that sometimes new information is absorbed into a stock's price in more than a single trading day can lead to an abnormal return for both individual and institutional traders. As shown in this research, the ability of the market participants such as analysts, and institutional and individual investors to evaluate the new public information in a short period of time is limited. Moreover, regulatory authorities must impose more regulations upon corporate information disclosure in both speed, accuracy, and depth that would enable market participants to evaluate quickly and accurately the correct economic impact of the new information on the firm's value.

Appendix A

The following tables show the p-values of the three tests we perform for every net daily gain and the win days' proportion within each strategy: T-test, Sign test and Wilcoxon test.

Note that p-values of the majority of the tests is effectively zero which is not surprising since even small deviations of the relevant statistics (i.e., sample mean or median) from the hypothesized value in the null hypothesis will lead to very small p-values when the sample size is relatively large. In our case hundreds of datapoints are used for each estimated parameter in each of the three statistical tests. On the other hand, the p-values are close to one in the scenarios when a statistic differs from the hypothesized value in the opposite direction of the alternative hypothesis (for example when the alternative hypothesis states that the population mean is greater than zero, however, the sample mean is smaller than zero).

Nasdaq

Strategy	Win Days proportion			Net Daily Gain		
	T-test	Sign-test	Wilcoxon test	T-test	Sign-test	Wilcoxon test
1	0.07	6.55 e-12	1.6 e-8	2.02 e-5	<2.2 e-16	2.48 e-16
2	0.00009	4.14 e-14	7.6 e-11	7.5 e-9	<2.2 e-16	<2.2 e-16
3	0.99	0.23	0.69	0.98	3.5 e-10	1.7 e-5
4	0.0003	2.2e-6	0.00004	0.02	7.08 e-5	0.0016
5	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16
6	2.34 e-15	<2.2 e-16	<2.2 e-16	1.03 e-11	<2.2 e-16	<2.2 e-16
7	1.4 e-7	<2.2 e-16	7.3 e-16	2.4 e-14	<2.2 e-16	<2.2 e-16
8	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16
9	4.97 e-13	5.41 e-16	<2.2 e-16	2.6 e-15	<2.2 e-16	<2.2 e-16
10	0.022	6.9 e-6	3.9 e-5	1.3 e-7	<2.2 e-16	4.3 e-12

S&P

Strategy	Win Days proportion			Net Daily Gain		
	T-test	Sign-test	Wilcoxon test	T-test	Sign-test	Wilcoxon test
1	3.7 e-6	<2.2 e-16	<2.2 e-16	8.9 e-16	<2.2 e-16	<2.2 e-16
2	4.5 e-7	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16
3	<2.2 e-16	3.8 e-6	0.002	1.9 e-8	<2.2 e-16	<2.2 e-16
4	<2.2 e-16	3.7 e-8	0.002	2.8 e-12	<2.2 e-16	<2.2 e-16
5	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16
6	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16
7	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16
8	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16
9	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16
10	5.9 e-8	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16

Russell 2000

Strategy	Win Days proportion			Net Daily Gain		
	T-test	Sign-test	Wilcoxon test	T-test	Sign-test	Wilcoxon test
1	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16
2	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16
3	0.99	2.8 e-10	0.002	0.99	<2.2 e-16	<2.2 e-16
4	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16
5	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16
6	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16
7	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16
8	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16
9	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16
10	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16	<2.2 e-16

Abbreviations

SEC Securities and Exchange Commission

Acknowledgements Not applicable.

Authors' Contributions The Authors declare equal contribution for this research paper.

Funding There is no funding source to declare.

Data Availability All Data and materials are available upon request.

Declarations

Competing Interests There is no competing interest to declare.

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